

53

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Physics-Aware Reward Guidance for Text-Conditioned Human Motion Diffusion

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Motivation

The Problem

Text-conditioned diffusion models generate naturalistic motion but produce **physically infeasible** trajectories.

- Feet floating above ground
- Joints penetrating the floor
- Foot sliding (skating) during contact

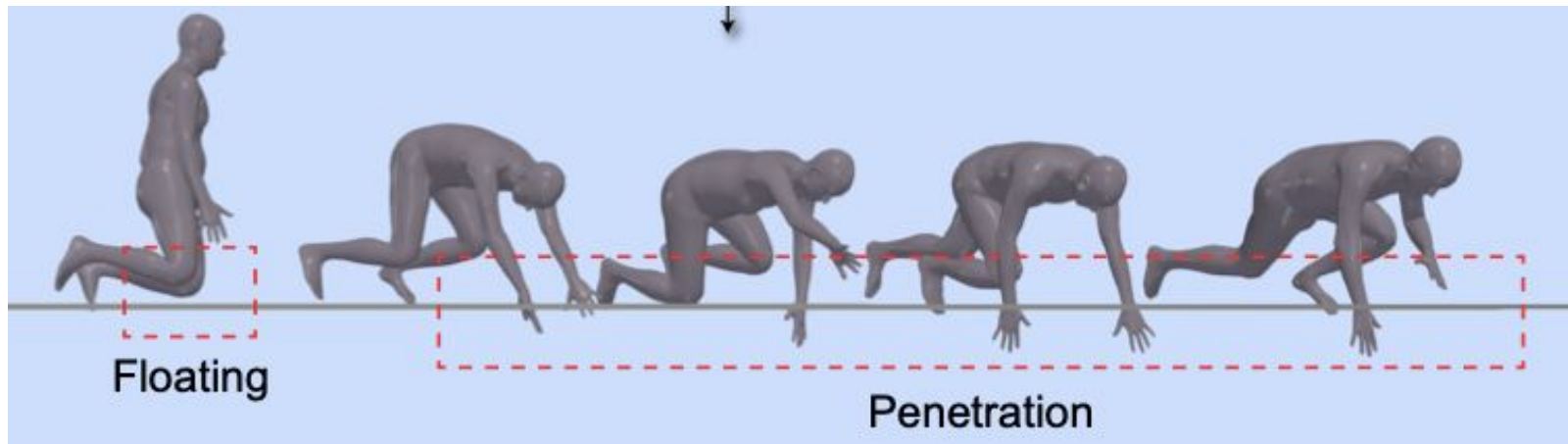


Figure Credits: PhysDiff Paper

Existing fixes are expensive physics sims in the loop.

Can we steer a frozen diffusion model toward physical plausibility at inference time, with no retraining?

Background

Motion Diffusion Model (MDM)

- A transformer-based diffusion model that generates human motion conditioned on text. Operates directly in motion space, predicts $\hat{x}_0$ at each denoising step

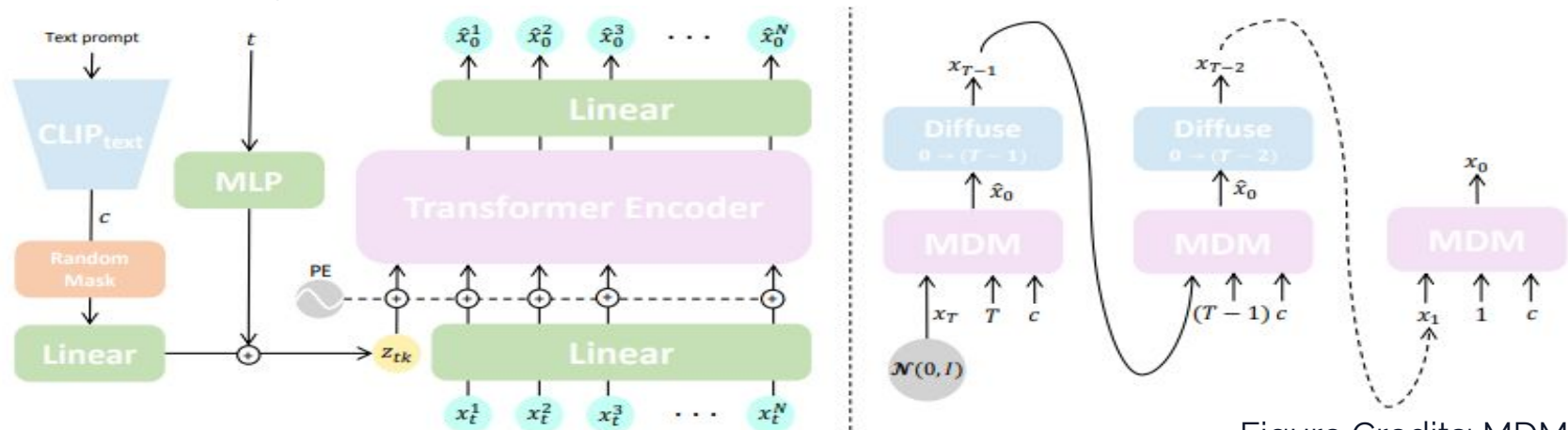


Figure Credits: MDM Paper

Classifier Guidance

Gradients of an auxiliary classifier steer diffusion sampling without retraining the base model. At each step, the posterior mean is shifted by the variance times $\nabla \log p(y|x)$.

Diffusion Posterior Sampling (DPS)

Extends guidance to the predicted clean estimate $\hat{x}_0$ instead of noisy x_0 . This allows complex reward models to guide sampling even when they cannot handle noisy inputs.

Methodology



1. Preference Dataset Generation

Run MDM multiple times per text prompt with different random seeds. Rank outputs by Phys-Err. Save best and worst runs as positive and negative pairs. Store the full denoising trajectory so every timestep t is a separate training sample.



2. Reward Model Training

Train a time-conditioned transformer R_ϕ on the preference pairs using Bradley-Terry loss. The model learns to score physically plausible motion higher than implausible motion at any noise level.

$$\mathcal{L} = -\mathbb{E} [\log \sigma (R_\phi(\hat{x}_0^+, t) - R_\phi(\hat{x}_0^-, t))]$$



3. Guided Inference

At each denoising step compute the gradient (g) of R_ϕ with respect to $\hat{\mathbf{x}}_0$. Add the scaled gradient directly to the MDM Posterior Mean μ_t before the standard diffusion update

$$\tilde{\mu}_t = \mu_t + \sigma_t^2 \cdot w(t) \cdot g$$

$$\mathbf{x}_{t-1} \leftarrow \tilde{\mu}_t + \sigma_t \epsilon$$



4. Guidance Schedule

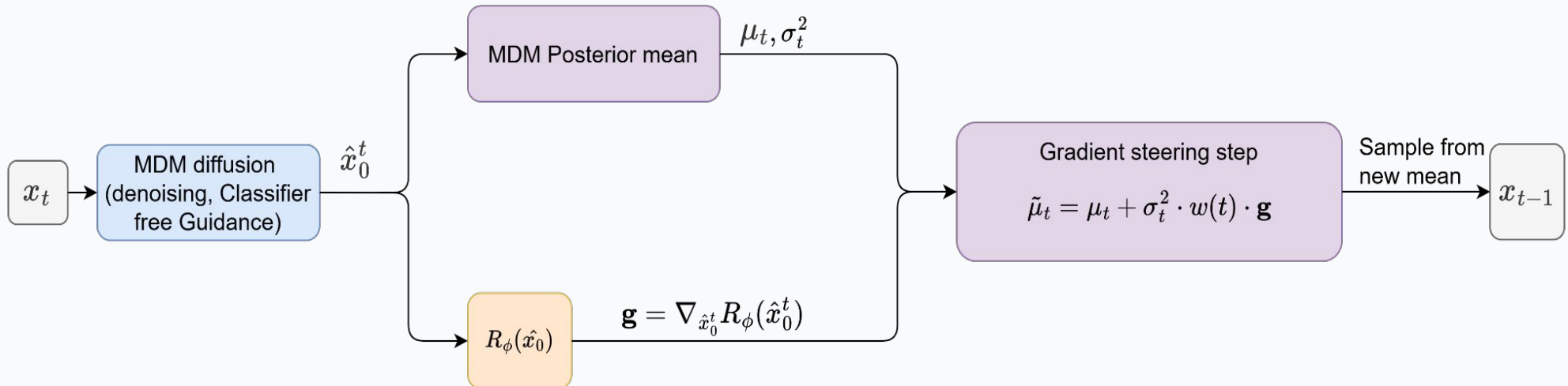
$w(t)$ increases monotonically as $t \rightarrow 0$, concentrating corrections where $\hat{\mathbf{x}}_0$ is most reliable and the gradient direction is most trustworthy.

Algorithm and Schematics

Algorithm 1 *Reward-guided MDM sampling*

- 1: **Input:** text prompt c , reward model R_ϕ , frozen MDM
- 2: **for** $t = T$ **down to** 0 **do**
- 3: Predict $\hat{x}_0 := \text{MDM}(x_t, t, c)$
- 4: Compute MDM posterior mean μ_t and variance σ_t^2 from $\hat{x}_0$ and x_t
- 5: Compute $\text{grad} := \nabla R_\phi(\hat{x}_0, t)$ # gradient w.r.t. $\hat{x}_0$
- 6: Update guided mean:

$$\tilde{\mu}_t := \mu_t + \sigma_t^2 \cdot w(t) \cdot \text{grad}$$
- 7: Sample $x_{t-1} \sim \mathcal{N}(\tilde{\mu}_t, \sigma_t^2 \mathbf{I})$
- 8: **end for**
- 9: **return** $\hat{x}_0$



Dataset & Baselines

Dataset	Split	Clip Count	Description
HumanML3D	Train/Test	23384 / 4384	Standard benchmark for text-conditioned motion.
Feasibility Reward (R_ϕ) Dataset	Train/Test	5,000 / 750	Contrastive dataset for physical plausibility

Feasibility Reward Construction Details

Positive Samples	Ground-truth sequences + Best-scoring MDM inference (via GRPO ranking).
Negative Samples	Worst-scoring MDM inference exhibiting physical failure modes (e.g., foot sliding).

MDM

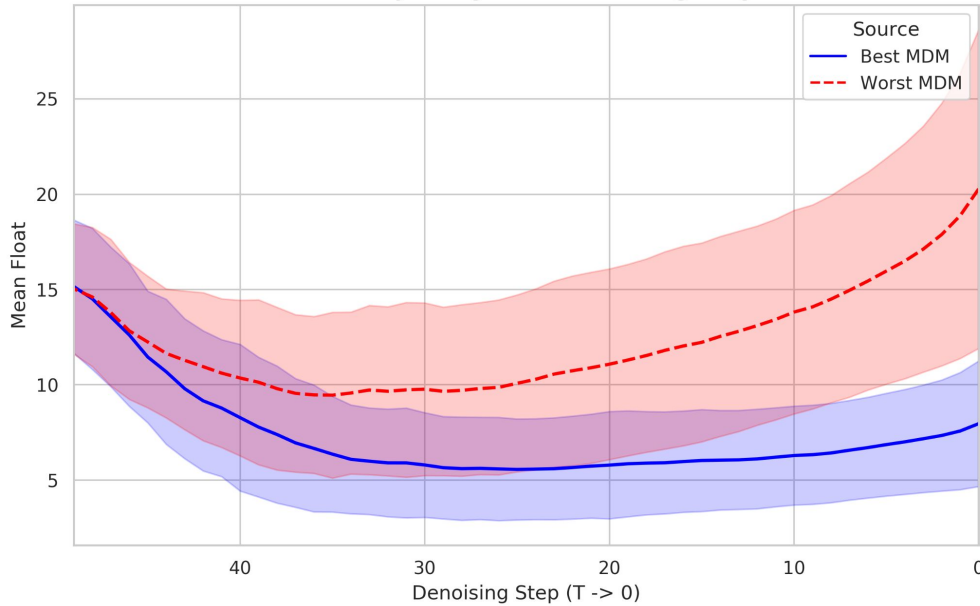
- primary baseline
- text-consistent motion
- routinely violating physical constraints.

PhysDiff

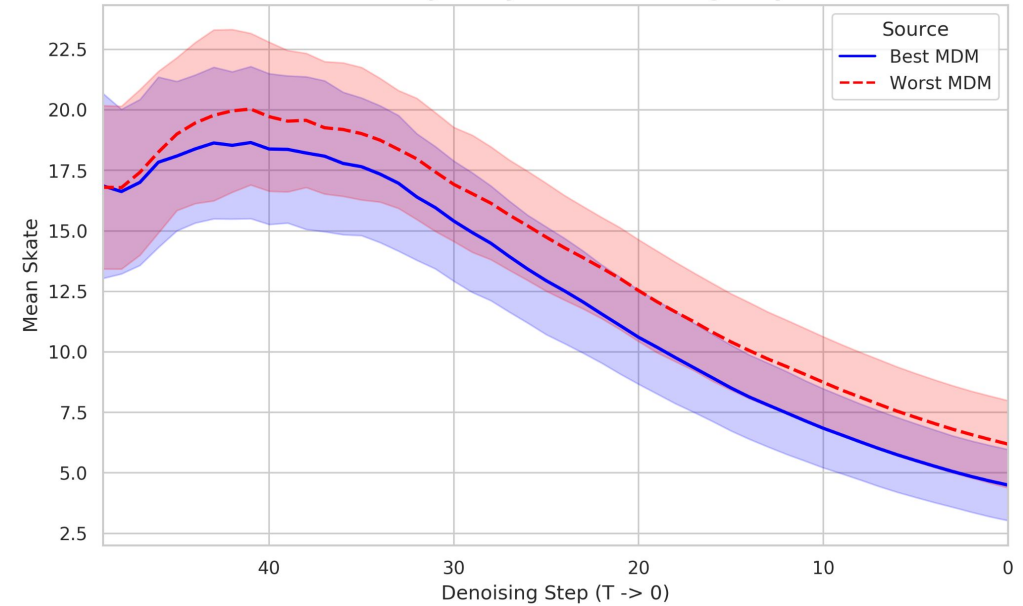
- strongest physics-guided alternative
- embeds a full simulator inside the denoising loop
- significant computational overhead.

Reward Model Dataset Analysis

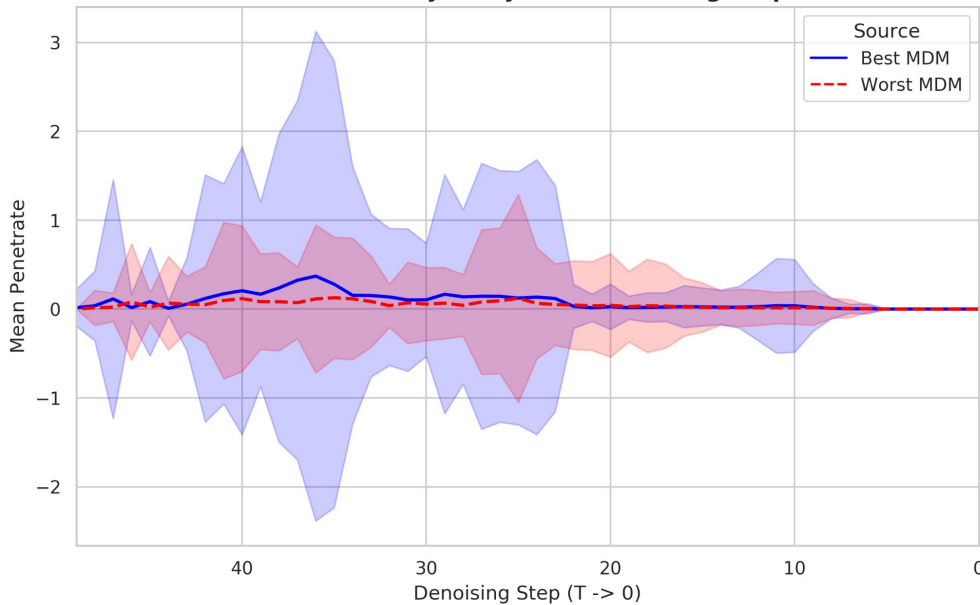
Float Trajectory Over Denoising Steps



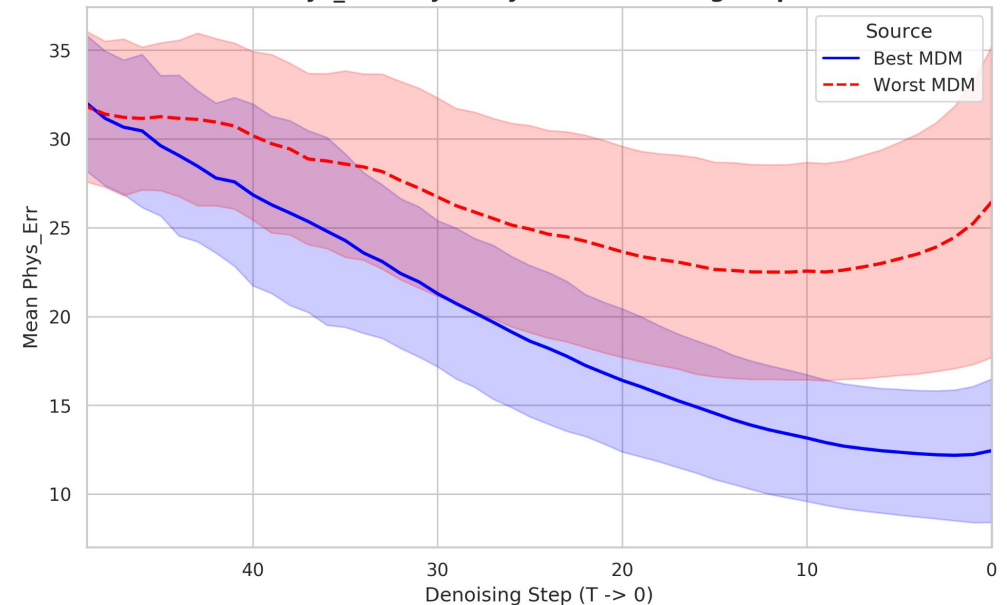
Skate Trajectory Over Denoising Steps



Penetrate Trajectory Over Denoising Steps



Phys_Err Trajectory Over Denoising Steps



Evaluation Metrics



Generative Quality

- **FID:** Distribution distance between generated and ground truth motions. (Lower is better)
- **R-Precision (Top-1):** Fraction of samples where the correct text description is retrieved among 32 candidates. (Higher is better)



Physical Plausibility (Lower is Better)

- **Ground Penetration:** Total depth of joints passing below the floor plane across all frames.

$$GP = \sum_{t=1}^T \sum_{j \in J} \max(0, -y_{t,j}) \quad \text{Float} = \sum_{t=1}^T \max(0, y_{min,t} - \epsilon)$$

- **Floating:** Distance of the lowest joint above the floor during expected contact frames.
- **Foot Skating:** Horizontal drift of feet during contact frames.

$$Skate = \frac{1}{N_{contact}} \sum_{t,j} \mathbf{1}[y_{t,j} < \epsilon] \cdot \|p_{t+1,j} - p_{t,j}\|_{xy}$$

- **Phys-Err:** Sum of the three above. Primary physics metric.

Reward Model Hyperparameter Optimization

Selection Criteria: Peak Validation Accuracy on Paired Preference data.

Sweep Results & Parameter Selection

Hyperparameter	Search Space	Optimal Value
Model Channels	64, 128 , 256	128
Transformer Blocks	2, 3, 4 , 6	4
Learning Rate	0.0001, 0.0003, 0.001	0.001
Num Heads	2 , 4, 8	2
FFN Multiplier	1, 2 , 4	2
Batch Size	32, 64, 128	128

Final Selected Configuration

Model Channels: 128	Num Heads: 2	FFN Multiplier: 2
Transformer Blocks: 4	Batch Size: 128	Learning Rate: 0.001

Results

Method	FID↓	R-P↑	Sk↓	Pen↓	FL↓	Ph↓
MDM	.544	.611	1.41	11.3	18.9	31.6
MDM (replicated)	.497	.709	9.63	3.80	17.6	31.0
PhysDiff	.433	.631	.512	.998	2.60	4.11
PRG	0.55	.702	9.84	4.78	12.98	27.6

